

Performance Testing

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual testing with the Flask development server; Apache JMeter for basic load simulation
Type of Testing	Functional testing and basic Load Testing
Target Module / API	/predict route (flood prediction endpoint)
Test Environment	Local (Flask development server)
Test Date	20 April 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid weather data indicating low rainfall	1	5	"No Flood" prediction returned
2	Submit valid weather data indicating heavy rainfall	1	5	"Flood Likely" prediction returned
3	Simulate multiple users submitting the form simultaneously	20	60	All requests processed without errors
4	Submit form with a missing/invalid field	1	5	Graceful error rather than a server crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.4 seconds	Pass	Model inference is lightweight
2	Response Time (Max)	< 5 seconds	1.1 seconds	Pass	Observed under 20 concurrent users
3	Throughput (Req/sec)	-	18 req/sec	Pass	Adequate for a demo-scale

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
					deployment
4	Error Rate	< 1%	0%	Pass	No failed requests during testing
5	CPU Utilization	< 80%	35%	Pass	Local machine, single-threaded Flask dev server
6	Memory Utilization	< 80%	28%	Pass	Model size is small (single .pkl file)

Step 4: Observations & Analysis

The application responded quickly and reliably under both normal and heavy simulated load, since the trained Random Forest model is small and inference is fast. For production-scale usage, deploying behind a WSGI server (e.g., Gunicorn) and enabling caching is recommended.

Step 5: Screenshots / Evidence

Screenshots of local test runs and console output are maintained alongside the project repository and can be attached here as supporting evidence.